Machine Learning Project Report

Breast Cancer Diagnosis using Classification Algorithms

1. Problem statement

The objective is to classify a breast mass as benign or malignant from numeric measurements computed from digitized images of fine needle aspirate samples. This project demonstrates a standard supervised machine-learning workflow.

2. Dataset

The project uses the Wisconsin Diagnostic Breast Cancer dataset distributed through scikit-learn. It contains 569 observations, 30 numeric input features and a binary diagnosis target. The CSV included in this ZIP uses the labels 'benign' and 'malignant'.

3. Preprocessing methodology

• Missing values: inspected with isnull(); median imputation is included in the model pipelines for robustness. The provided dataset has no missing values.

• Categorical encoding: the diagnosis target is encoded as 0 = benign and 1 = malignant. Predictor variables are numeric.

• Feature scaling: StandardScaler is applied to Logistic Regression; tree-based models do not require scaling.

• Outliers: boxplots are used for inspection. No automatic removal is performed because extreme measurements may carry useful information.

• Feature selection: SelectKBest with ANOVA F-test selects the 15 highest-scoring features within each training pipeline.

• Imbalance: class counts are inspected; Random Forest uses class_weight='balanced'.

• Data split: stratified 80% training and 20% testing, with random_state=42. Preprocessing is fitted only on training data.

4. Algorithms

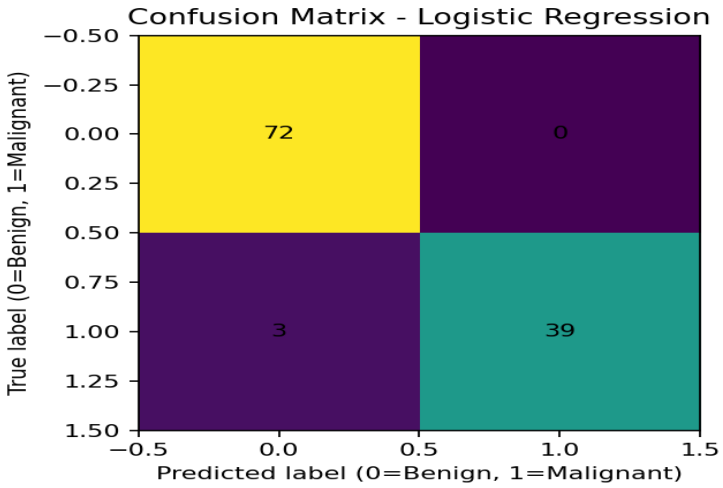
Logistic Regression, Random Forest Classifier and Gradient Boosting Classifier were selected as three suitable binary classification algorithms. Each was trained using the same training/test split.

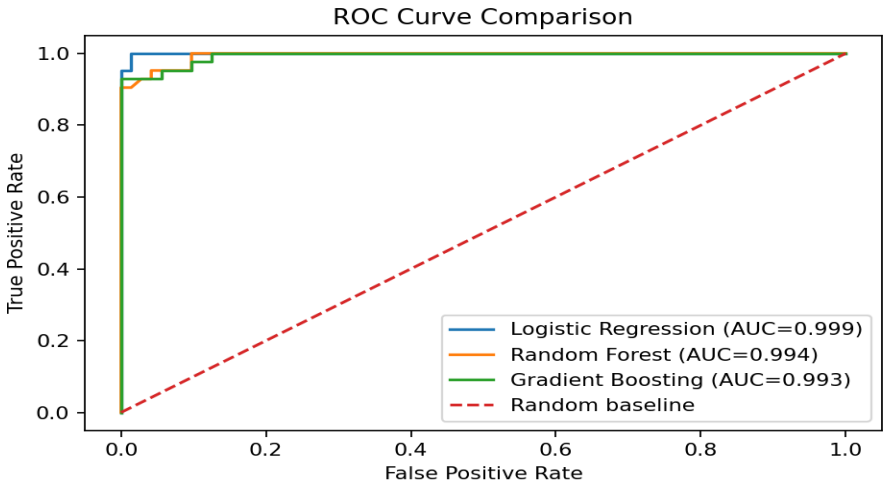
5. Evaluation and results

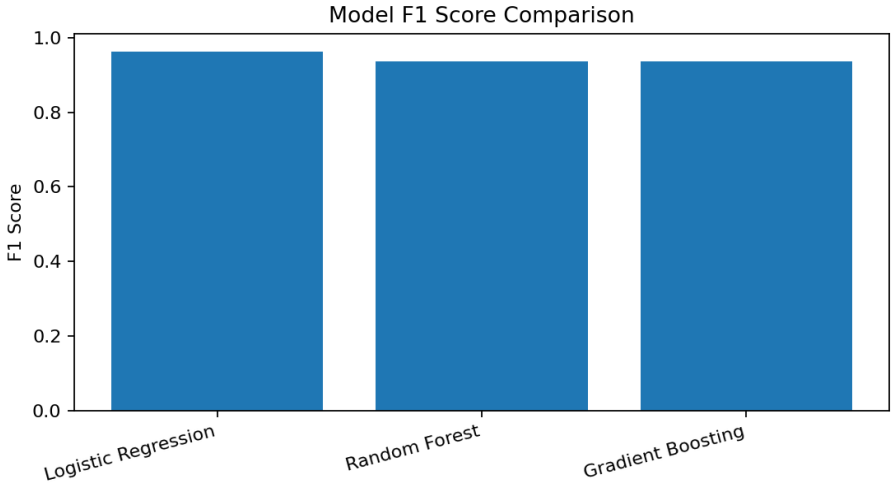
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.974	1.000	0.929	0.963	0.999
Random Forest	0.956	1.000	0.881	0.937	0.994
Gradient Boosting	0.956	1.000	0.881	0.937	0.993

The highest F1 score on this fixed test split was achieved by **Logistic Regression**. The included notebook prints the detailed classification report and displays the confusion matrix and ROC curves.

6. Visual insights







7. Conclusion and limitations

The experiment compares three classification models using accuracy, precision, recall, F1 score and ROC-AUC. The best model should be judged using the metric most relevant to the problem, not accuracy alone. Results depend on the random split and model settings. This project is strictly educational and must not be used to diagnose patients or guide medical treatment.

8. Dataset acknowledgement

Dataset: Wisconsin Diagnostic Breast Cancer dataset, available through the scikit-learn datasets module. See the scikit-learn documentation for dataset details and original references.